7. An Algorithmic Approach to Identification

A consolidated, mathematical lecture companion

English notes based on the lecture slides *An Algorithmic Approach to Identification* (Prof. Sanghack Lee, GSDS SNU)

The central question is how to decide identifiability of $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ mechanically, without searching for a clever sequence of do-calculus steps.

Roadmap. (1) Define a factorization of the distributions generated by an SCM based on its causal diagram (c-components and c-factors).

(2) Establish operations that identify particular factors from a distribution.

(3) Express the target effect in terms of factors and identify each one independently. These notes consolidate the slides, verify every example by explicit sums, and state the resulting algorithm with its completeness theorem. Physical PDF page numbers are listed in Section 5.

Contents

1 Factorizing observational distributions **2**

1.1 Confounded components . . . 2
1.2 Markovian case . . . 2
1.3 Semi-Markovian case . . . 2
1.4 C-factors . . . 3
1.5 C-factors are causal effects . . . 3

2 C-factor algebra **3**

2.1 Marginalizing variables in c-factors . . . 3
2.2 C-component factorization . . . 4
2.3 Visualization . . . 5

3 Causal queries in terms of c-factors: examples **5**

3.1 Back-door . . . 5
3.2 Front-door . . . 5
3.3 Napkin . . . 5
3.4 Overview of the three derivations . . . 6

4 A general identification algorithm **6**

4.1 Reducing the query to c-factors . . . 6
4.2 The Identify procedure . . . 7

5 Source coverage **7**

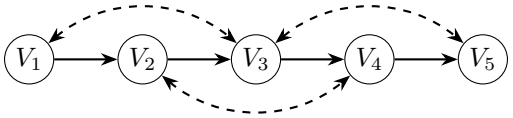
1 Factorizing observational distributions

Purpose. Group variables that share latent confounders and show that the joint distribution factorizes over these groups.

1.1 Confounded components

Definition 1.1 (C-component)**.** When V_i and V_j share a common unobservable parent U, their mechanisms are tied together through U, and V_i, V_j are in the same *confounded component* (c-component). The relation is closed under chaining: two variables are in the same c-component iff they are connected by a path of bidirected edges.

Running example: $V_1 \to V_2 \to V_3 \to V_4 \to V_5$ with U_1: $V_1 \leftrightarrow V_3$, U_2: $V_3 \leftrightarrow V_5$, U_3: $V_2 \leftrightarrow V_4$.



V_1 is in the same c-component as V_3, V_3 as V_5, hence V_1 as V_5; V_2 is with V_4. To see this easily, look at the graph induced by the bidirected edges only. The c-component relation is reflexive, symmetric and transitive, so it *partitions* the observed variables: here $\{V_1, V_3, V_5\}$ and $\{V_2, V_4\}$.

1.2 Markovian case

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

For $Z \to X$, $Z \to Y$, $X \to Y$:

$$\begin{aligned} P(z, x, y) &= \sum_{u_z} P(z \mid u_z) P(u_z) \sum_{u_x} P(x \mid z, u_x) P(u_x) \sum_{u_y} P(y \mid x, z, u_y) P(u_y) \\ &= P(z) P(x \mid z) P(y \mid x, z). \end{aligned}$$

The factors $P(v_i \mid \mathbf{pa}_i)$ are "canonical", and every one of them is computable from $P(\mathbf{v})$:

$$P(v_i \mid \mathbf{pa}_i) = \frac{\sum_{\mathbf{v} \setminus (\{v_i\} \cup \mathbf{pa}_i)} P(\mathbf{v})}{\sum_{\mathbf{v} \setminus \mathbf{pa}_i} P(\mathbf{v})},$$

$$\text{e.g. } P(z) = \sum_{x,y} P(\mathbf{v}),\ P(x \mid z) = \frac{\sum_y P(\mathbf{v})}{\sum_{x,y} P(\mathbf{v})},\ P(y \mid x, z) = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})}.$$

1.3 Semi-Markovian case

With only U_1 ($V_1 \leftrightarrow V_3$) and U_2 ($V_3 \leftrightarrow V_5$):

$$P(\mathbf{v}) = P(v_2 \mid v_1) P(v_4 \mid v_3) \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2).$$

Adding U_3 ($V_2 \leftrightarrow V_4$):

$$\begin{aligned} P(\mathbf{v}) = &\Big[\sum_{u_3} P(v_2 \mid v_1, u_3) P(v_4 \mid v_3, u_3) P(u_3) \Big] \\ &\cdot \Big[\sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2) \Big]. \end{aligned}$$

The sums separate along c-components because different c-components share no latent variable.

1.4 C-factors

These factors are long to write in terms of $P(\mathbf{v}, \mathbf{u})$, but their structure follows from the diagram. Define for $C \subseteq \mathbf{V}$

$$Q[C](\mathbf{c}, \mathbf{pa}_C) = \sum_{\mathbf{u}(C)} P(\mathbf{u}(C)) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i),$$
$$\mathbf{U}(C) = \bigcup_{V_i \in C} \mathbf{U}_i.$$

Then

$$P(\mathbf{v}) = Q[V_2, V_4](v_2, v_4, v_1, v_3) \cdot Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4),$$

abbreviated $P(\mathbf{v}) = Q[V_1, V_3, V_5]\, Q[V_2, V_4]$, without naming $\mathbf{U}$ at all. Note $Q[\mathbf{V}] = P(\mathbf{v})$, and by convention $Q[\emptyset] = 1$.

1.5 C-factors are causal effects

For $C \subseteq \mathbf{V}$, consider the effect of all other variables on C. By the truncated factorization,

$$P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = \sum_{\mathbf{u}(C)} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}(C)) = Q[C],$$

since latent variables that are not parents of C sum out. *C-factors are interventional distributions*; this is the key link between factorization and identification. For example $Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4) = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2)$ is the distribution of (V_1, V_3, V_5) when $V_2 = v_2$ and $V_4 = v_4$ are fixed by intervention.

2 C-factor algebra

Purpose. Develop the two operations, marginalizing to ancestral sets and factorizing into c-components, that drive identification.

2.1 Marginalizing variables in c-factors

C-factors behave partly like probabilities. In $Q[V_1, V_3, V_5]$ each of V_1, V_3, V_5 appears in a single term (none is a parent of another member), so summing over v_3 gives

$$\sum_{v_3} Q[V_1, V_3, V_5] = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) \Big(\sum_{v_3} P(v_3 \mid v_2, u_1, u_2) \Big) P(v_5 \mid v_4, u_2) = Q[V_1, V_5].$$

In $Q[V_1, V_2, V_3]$, by contrast, V_1 is a parent of V_2. Summing over v_1:

$$\sum_{v_1} Q[V_1, V_2, V_3] = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) \sum_{v_1} P(v_1 \mid u_1) P(v_2 \mid v_1, u_3),$$

whereas

$$Q[V_2, V_3](v_2, v_3, v_1) = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) P(v_2 \mid v_1, u_3)$$

keeps v_1 as a fixed argument. The coupling of v_1 with u_1 (shared with V_3) inside the sum prevents the reduction, so $\sum_{v_1} Q[V_1, V_2, V_3] \neq Q[V_2, V_3]$ in general: one can only sum out variables that do not feed into the remaining ones.

Lemma 2.1 (Ancestral reduction)**.** *Let $W \subseteq C \subseteq \mathbf{V}$. If W is ancestral in $\mathcal{G}[C]$ (it contains all its ancestors within the subgraph induced by C), then*

$$Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C].$$

Proof idea: sum out the variables of $C \setminus W$ in reverse topological order; each one, when summed, appears only in its own term (it has no children left in C) and that term sums to one. For $C = \{V_1, V_2, V_3\}$ in $\mathcal{G}[C]$ ($V_1 \to V_2 \to V_3$): $\{V_1, V_2\}$ and $\{V_1\}$ are ancestral, $\{V_2, V_3\}$ is not.

2.2 C-component factorization

Lemma 2.2 (Tian 2002)**.** *For any $H \subseteq \mathbf{V}$, let $H_1, \dots, H_k$ be the c-components of $\mathcal{G}[H]$. Then*

$$Q[H] = \prod_{j=1}^{k} Q[H_j].$$

Moreover, let $V^{(1)} < \cdots < V^{(|H|)}$ be a topological order of $\mathcal{G}[H]$ and $H^{\le i}$ the variables of H up to and including $V^{(i)}$. Then each factor is computable from $Q[H]$:

$$Q[H_j] = \prod_{V^{(i)} \in H_j} \frac{Q[H^{\le i}]}{Q[H^{\le i-1}]}, \qquad Q[H^{\le i}] = \sum_{\mathbf{h}\setminus\mathbf{h}^{\le i}} Q[H].$$

($H^{\le i}$ is ancestral in $\mathcal{G}[H]$, so $Q[H^{\le i}]$ follows by ancestral reduction.)

Example 2.3 (Factorizing $P(\mathbf{v})$ by c-components)**.** Graph: $Z_2 \to X$, $Z_2 \to Z_1$, $Z_2 \to Z_3$, $X \to Z_1$, $Z_1 \to Y$, $Z_3 \to Y$, $Z_2 \leftrightarrow X$, $X \leftrightarrow Z_3$, $X \leftrightarrow Y$, $Z_2 \leftrightarrow Y$. The c-components are $\{Z_2, Z_3, X, Y\}$ and $\{Z_1\}$, so

$$Q[\mathbf{V}] = Q[Z_2, Z_3, X, Y]\, Q[Z_1],$$
$$P(\mathbf{v}) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, z_3, x, y)) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, x)),$$

the last step because only the parents Z_2, X of Z_1 matter (Rule 3).

Example 2.4 (Running example)**.** With $H = \mathbf{V}$ and the order $V_1 < \cdots < V_5$, $Q[\mathbf{V}] = Q[V_1, V_3, V_5]Q[V_2, V_4]$ and

$$\begin{aligned}
Q[V_1, V_3, V_5] &= \frac{Q[V_1]}{Q[\emptyset]} \cdot \frac{Q[V_1, V_2, V_3]}{Q[V_1, V_2]} \cdot \frac{Q[V_1, \dots, V_5]}{Q[V_1, \dots, V_4]} \\
&= \frac{\sum_{v_2,\dots,v_5} P(\mathbf{v})}{\sum_{\mathbf{v}} P(\mathbf{v})} \cdot \frac{\sum_{v_4,v_5} P(\mathbf{v})}{\sum_{v_3,v_4,v_5} P(\mathbf{v})} \cdot \frac{P(\mathbf{v})}{\sum_{v_5} P(\mathbf{v})} \\
&= \frac{P(v_1)}{1} \cdot \frac{P(v_1, v_2, v_3)}{P(v_1, v_2)} \cdot \frac{P(\mathbf{v})}{P(v_1, \dots, v_4)} = P(v_1)P(v_3 \mid v_1, v_2)P(v_5 \mid v_1, \dots, v_4), \\
Q[V_2, V_4] &= \frac{Q[V_1, V_2]}{Q[V_1]} \cdot \frac{Q[V_1, \dots, V_4]}{Q[V_1, V_2, V_3]} = P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).
\end{aligned}$$

In $\mathcal{G}[\{V_2, V_4\}]$ there are no directed edges ($V_2 \leftrightarrow V_4$ only), so both $\{V_2\}$ and $\{V_4\}$ are ancestral there:

$$Q[V_2] = \sum_{v_4} Q[V_2, V_4] = P(v_2 \mid v_1) \sum_{v_4} P(v_4 \mid v_1, v_2, v_3) = P(v_2 \mid v_1),$$
$$Q[V_4] = \sum_{v_2} P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).$$

2.3 Visualization

Starting from $Q[\mathbf{V}]$ one can move down a lattice of c-factors: summing out a variable that is "last" (ancestral reduction, e.g. $Q[\mathbf{V}] \to Q[V_{1234}] \to Q[V_{123}] \to Q[V_{12}] \to Q[V_1]$) or splitting along c-components (e.g. $Q[\mathbf{V}] \to Q[V_{135}] \cdot Q[V_{24}]$, $Q[V_{1234}] \to Q[V_{13}] \cdot Q[V_{24}]$, $Q[V_{123}] \to Q[V_{13}] \cdot Q[V_2]$). From $Q[V_{135}]$ every subset is reachable by summation since $\mathcal{G}[\{V_1, V_3, V_5\}]$ has no directed edges, and $Q[V_{15}] = Q[V_1]Q[V_5]$ factorizes because V_1, V_5 are not bidirected-connected inside $\{V_1, V_5\}$. Once a factor is fully factorized it cannot be factorized further; summation is commutative.

Summary of the algebra.

1. *Reduce to an ancestral set:* $Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C]$ if W is ancestral in $\mathcal{G}[C]$.
2. *Factorize into c-components:* $Q[H] = \prod_j Q[H_j]$, with $H_1, \ldots, H_k$ the c-components of $\mathcal{G}[H]$, each computable from $Q[H]$.

These two operations yield an if-and-only-if condition for identifiability.

3 Causal queries in terms of c-factors: examples

Purpose. Rederive the back-door, front-door and napkin formulas with c-factor algebra.

3.1 Back-door

$Z \to X$, $Z \to Y$, $X \to Y$ (Markovian): $P(\mathbf{v}) = Q[X]Q[Y]Q[Z]$. In $\mathcal{G}_{\overline{X}}$, $P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$ with

$$Q[Y] = \frac{Q[X,Y,Z]}{Q[X,Z]} = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})} = P(y \mid x, z), \qquad Q[Z] = \sum_{x,y} Q[X,Y,Z] = P(z),$$

so $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid z, x)P(z)$.

3.2 Front-door

$X \to Z \to Y$, $X \leftrightarrow Y$: $P(\mathbf{v}) = Q[X,Y] \cdot Q[Z]$ with (order $X < Z < Y$)

$$Q[X,Y] = \frac{Q[X]}{Q[\emptyset]} \cdot \frac{Q[X,Z,Y]}{Q[X,Z]} = P(x)P(y \mid x, z), \qquad Q[Z] = \frac{Q[X,Z]}{Q[X]} = P(z \mid x).$$

Then

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y, z \mid \mathrm{do}(x)) = \sum_z Q[Y,Z] = \sum_z Q[Y]Q[Z],$$

and since $\{Y\}$ is ancestral in $\mathcal{G}[\{X,Y\}]$ (no directed edge between them), $Q[Y] = \sum_{x'} Q[X,Y] = \sum_{x'} P(x')P(y \mid x', z)$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(x')P(y \mid x', z)$.

3.3 Napkin

$W_1 \to W_2 \to X \to Y$, $W_1 \leftrightarrow X$, $W_1 \leftrightarrow Y$. Then $P(\mathbf{v}) = Q[W_1, X, Y]\, Q[W_2]$ and

$$P(y \mid \mathrm{do}(x)) = \sum_{w_1, w_2} Q[W_1, W_2, Y] = Q[Y],$$

because $\{Y\}$ is ancestral in $\mathcal{G}[\{W_1, W_2, Y\}]$ (in that subgraph Y has no parents). Can we get $Q[Y]$ from $Q[W_1, X, Y]$, which is computable from $Q[\mathbf{V}]$? With the order $W_1 < W_2 < X < Y$,

$$Q[W_1, X, Y] = \frac{Q[W_1]}{Q[\emptyset]} \cdot \frac{Q[W_1, W_2, X]}{Q[W_1, W_2]} \cdot \frac{Q[W_1, W_2, X, Y]}{Q[W_1, W_2, X]} = P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x).$$

$\{X, Y\}$ is ancestral in $\mathcal{G}[\{W_1, X, Y\}]$, so $Q[X, Y] = \sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)$. In $\mathcal{G}[\{X, Y\}]$, X and Y are no longer bidirected-connected (W_1 is gone), so $Q[X, Y] = Q[X]Q[Y]$, giving

$$P(y \mid \mathrm{do}(x)) = Q[Y] = \frac{Q[X, Y]}{Q[X]} = \frac{Q[X, Y]}{\sum_y Q[X, Y]} = \frac{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)}{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)}.$$

The same result by do-calculus.

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x, w_1, w_2)) && \text{Rule 3} \\
&= P(y \mid x, \mathrm{do}(w_1, w_2)) && \text{Rule 2} \\
&= \frac{P(y, x \mid \mathrm{do}(w_1, w_2))}{P(x \mid \mathrm{do}(w_1, w_2))} = \frac{P(y, x \mid \mathrm{do}(w_2))}{P(x \mid \mathrm{do}(w_2))} && \text{Rule 3} \\
&= \frac{\sum_{w_1} P(y, x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))}{\sum_{w_1} P(x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))} && \\
&= \frac{\sum_{w_1} P(y, x \mid w_2, w_1)P(w_1)}{\sum_{w_1} P(x \mid w_2, w_1)P(w_1)} && \text{Rules 2 and 3.}
\end{aligned}$$

3.4 Overview of the three derivations

Back-door	Front-door	Napkin
$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X, Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[W_1, X, Y]\,Q[W_2]$
	$Q[X, Y] \xrightarrow{\text{sum}} Q[Y]$	$Q[W_1, X, Y] \xrightarrow{\text{sum}} Q[X, Y] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]$
$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = Q[Y]$

4 A general identification algorithm

Purpose. Combine the two operations into a sound and complete procedure.

4.1 Reducing the query to c-factors

Given $\mathcal{G}$ and $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, first use Rule 3 to enlarge the intervention: every variable that is not an ancestor of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$ (non-ancestors of $\mathbf{Y}$, and variables that affect $\mathbf{Y}$ only through $\mathbf{X}$) can be intervened on without changing the query. Let $C = An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} Q[\mathbf{V} \setminus \mathbf{X}] = \sum_{\mathbf{c} \setminus \mathbf{y}} Q[C].$$

If $\mathcal{G}[C]$ has c-components $C_1, \ldots, C_k$, then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} \prod_{i=1}^{k} Q[C_i],$$

while the data give $P(\mathbf{v}) = \prod_j Q[T_j]$ with $T_1, \ldots, T_\ell$ the c-components of $\mathcal{G}$, each $Q[T_j]$ computable from $P(\mathbf{v})$. Each C_i lies inside a single T_j. It remains to decide whether $Q[C_i]$ is computable from $Q[T_j]$.

4.2 The Identify procedure

Algorithm 1 IDENTIFY$(C, T, Q[T], \mathcal{G})$ (decides whether $Q[C]$ is computable from $Q[T]$, $C \subseteq T$)

```
A ← An(C)_{G[T]}
if A = C then                                   ▷ C is ancestral in T
    return Q[C] = Σ_{t\c} Q[T]                  ▷ marginalize
else if A = T then                              ▷ nothing to marginalize, unable to factorize
    return FAIL
else
    A_i ← the c-component of G[A] containing C
    obtain Q[A] = Σ_{t\a} Q[T], then Q[A_i] by c-component factorization   ▷ marginalize, then factorize
    return IDENTIFY(C, A_i, Q[A_i], G)
```

The three cases. (1) $A = C$: marginalize, $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$.

(2) $A = T$: C is strictly inside T, but T is its own ancestral closure and a single c-component, so T cannot be shrunk: fail.

(3) $C \subsetneq A \subsetneq T$: marginalize to A, factorize A into c-components $A_i, A_j, \ldots$, keep the one containing C (it must be contained in a single component because C is a c-component of $\mathcal{G}[C]$ and $C \subseteq A_i$), and recurse. Each step strictly shrinks T (recursive shrinking: $T \supsetneq A_i \supsetneq \cdots$), so the procedure terminates in at most $|T|$ steps.

Theorem 4.1 (Completeness; Tian 2002)**.** *$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $\mathcal{G}$ and $P(\mathbf{v})$ if and only if each c-factor $Q[C_i]$ is identified by* IDENTIFY$(C_i, T_i, Q[T_i], \mathcal{G})$*, where T_i is the c-component of $\mathcal{G}$ containing C_i.*

See also Huang and Valtorta (2006) and Shpitser and Pearl (2006), who proved completeness of the equivalent ID algorithm and, through it, of do-calculus. The failure case $A = T$ corresponds to a graphical structure (a "hedge") from which two models agreeing on $P(\mathbf{v})$ but not on the query can be built.

Example 4.2 (Bow graph fails)**.** $X \to Y$, $X \leftrightarrow Y$, query $P(y \mid \mathrm{do}(x))$: $C = An(Y)_{\mathcal{G}_{\overline{X}}} = \{Y\}$, $T = \{X, Y\}$. IDENTIFY$(\{Y\}, \{X, Y\}, \ldots)$: $A = An(Y)_{\mathcal{G}[T]} = \{X, Y\} = T$, so it fails, matching Lecture 4.

Example 4.3 (Front-door succeeds)**.** $X \to Z \to Y$, $X \leftrightarrow Y$: $C = \{Z, Y\}$ with c-components $\{Z\}$, $\{Y\}$ in $\mathcal{G}[C]$. $Q[Z]$ is itself a c-component of $\mathcal{G}$. For $Q[Y]$ with $T = \{X, Y\}$: $A = An(Y)_{\mathcal{G}[T]} = \{Y\} = C$ (the edge $X \to Z \to Y$ is not inside T), so $Q[Y] = \sum_x Q[X, Y]$, as found above.

5 Source coverage

The source deck has 91 physical pages (41 numbered slides with build-ups).

Topic	Physical pages
Roadmap, c-components, Markovian and semi-Markovian factorization	1–15
C-factors, causal-effect interpretation, marginalization	16–27
C-component factorization and examples	28–43
Visualization and summary of the algebra	44–54
Back-door, front-door, napkin via c-factors and do-calculus	55–76
General algorithm, Identify, completeness, visualization, references	77–91